

Lecture 13: Data Visualization Case Study

Heather Mattie, PhD

October 14, 2025

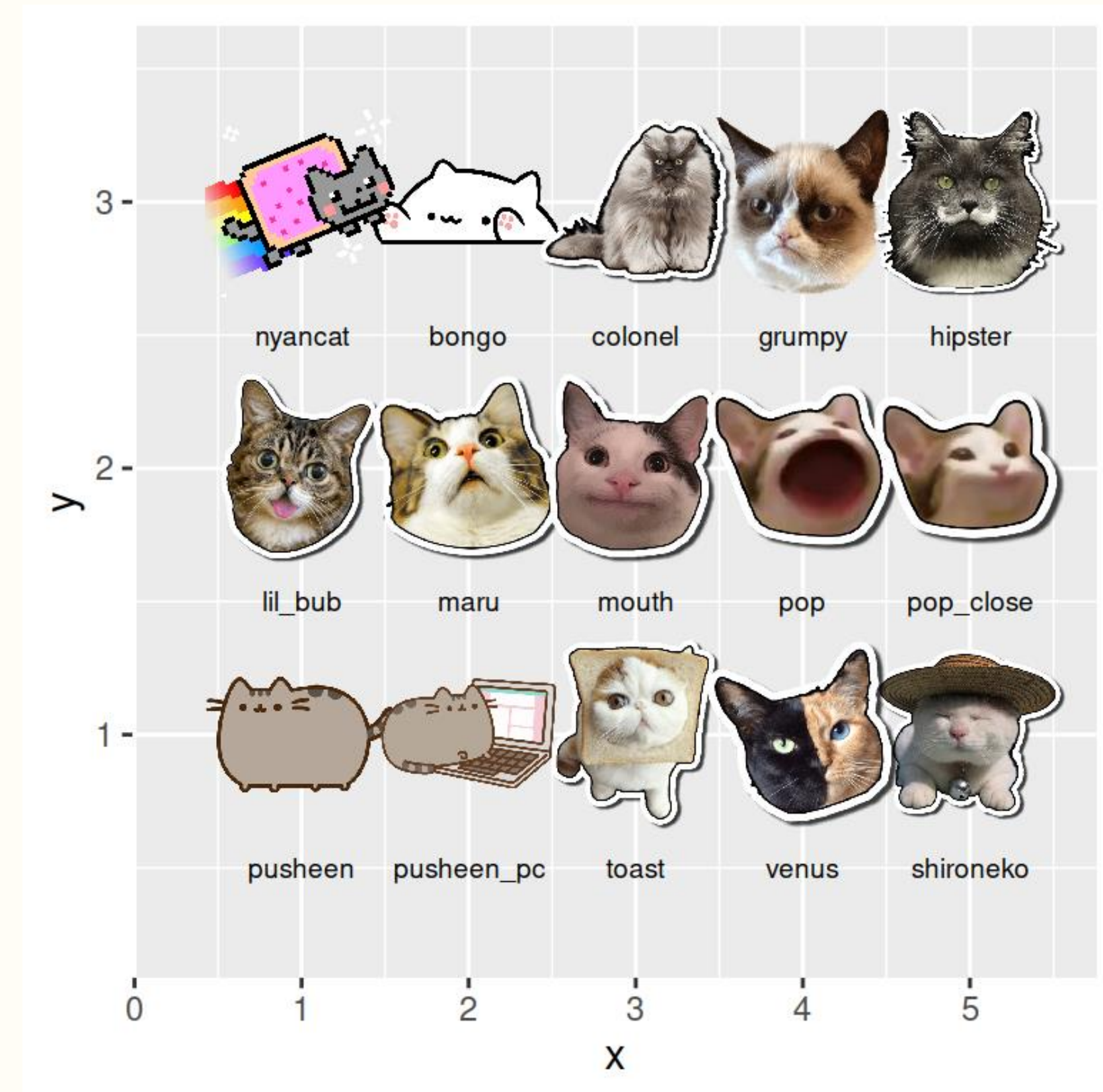
Recipe of the Day!

Spaghetti Carbonara



Announcements

- Homework #3 due Friday, October 24 by 11:59pm
- Lab will be an extended office hour this week
- Mid-course evaluation is live!



ggcats and geom_cat

Coding question of the day

Using the `murders` dataset from the `dslabs` package and the `state.x77` dataset that is built in to R, make a scatterplot with the total number of gun murders (`total`) on the y-axis and land area in square miles (`Area`) on the x-axis. You will need to use the `left_join` function to join the `state.x77` dataset to the `murders` dataset using the `state` column.

Bonus challenge: are there any NAs in the merged dataset? How are they created?

Be sure to run this code first

```
# Load necessary libraries
library(dplyr)
library(dslabs)
library(ggplot2)

# Load the state.x77 dataset and convert to a data frame
data("state")
state_data <- as.data.frame(state.x77)

# Add the state names to the state.x77 dataset
state_data$state <- rownames(state_data)

head(state_data)

# Load the murders dataset from the dslabs package
data("murders")
```

Comparing Distributions

Comparing Distributions Roadmap

1. Reordering
2. Smooth density plots

Comparing Distributions

Reordering

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Gapminder regions

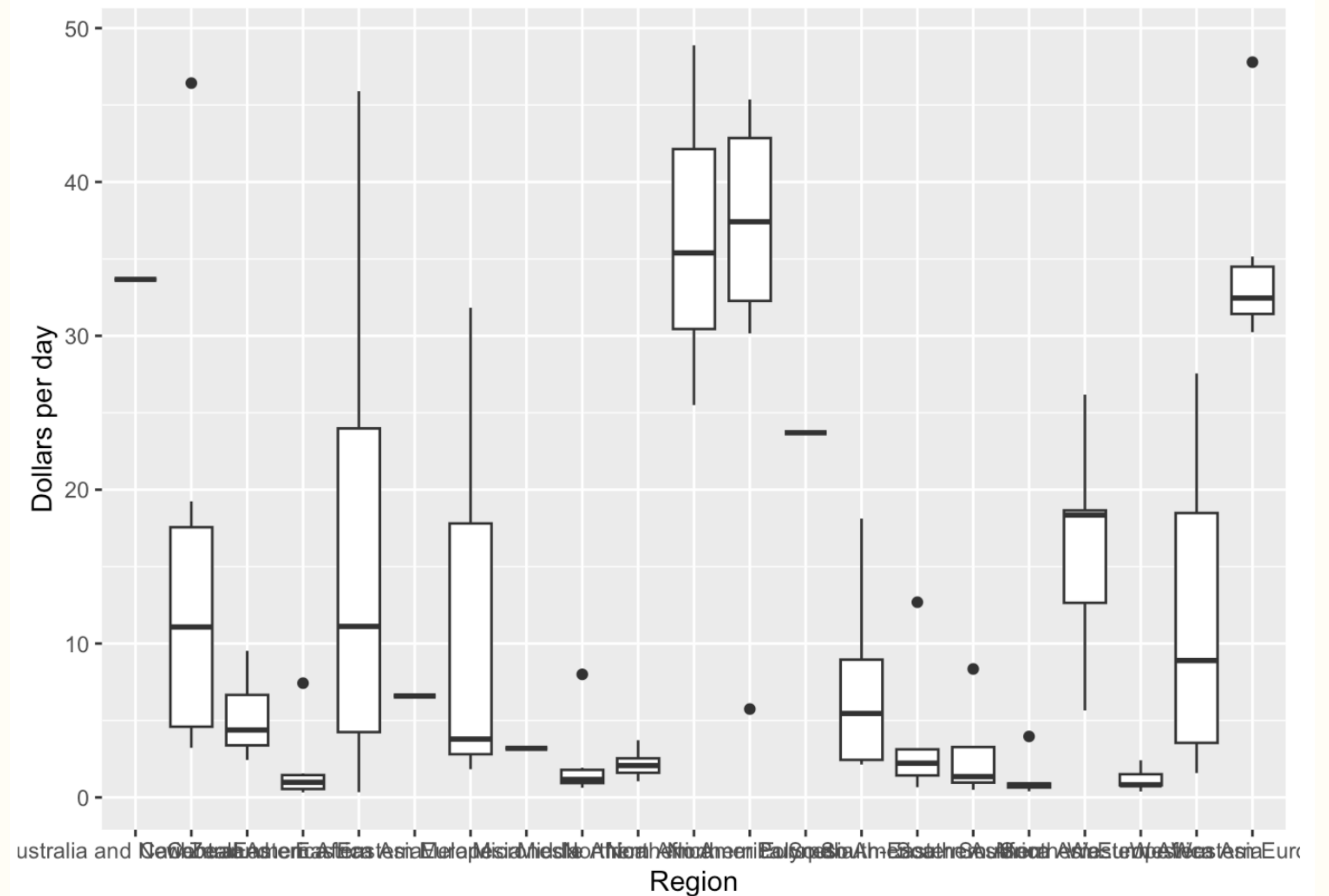
```
levels(gapminder$region)
```

```
## [1] "Australia and New Zealand" "Caribbean"
## [3] "Central America"          "Central Asia"
## [5] "Eastern Africa"           "Eastern Asia"
## [7] "Eastern Europe"           "Melanesia"
## [9] "Micronesia"               "Middle Africa"
## [11] "Northern Africa"          "Northern America"
## [13] "Northern Europe"          "Polynesia"
## [15] "South America"            "South-Eastern Asia"
## [17] "Southern Africa"          "Southern Asia"
## [19] "Southern Europe"          "Western Africa"
## [21] "Western Asia"             "Western Europe"
```

```
length(levels(gapminder$region))
```

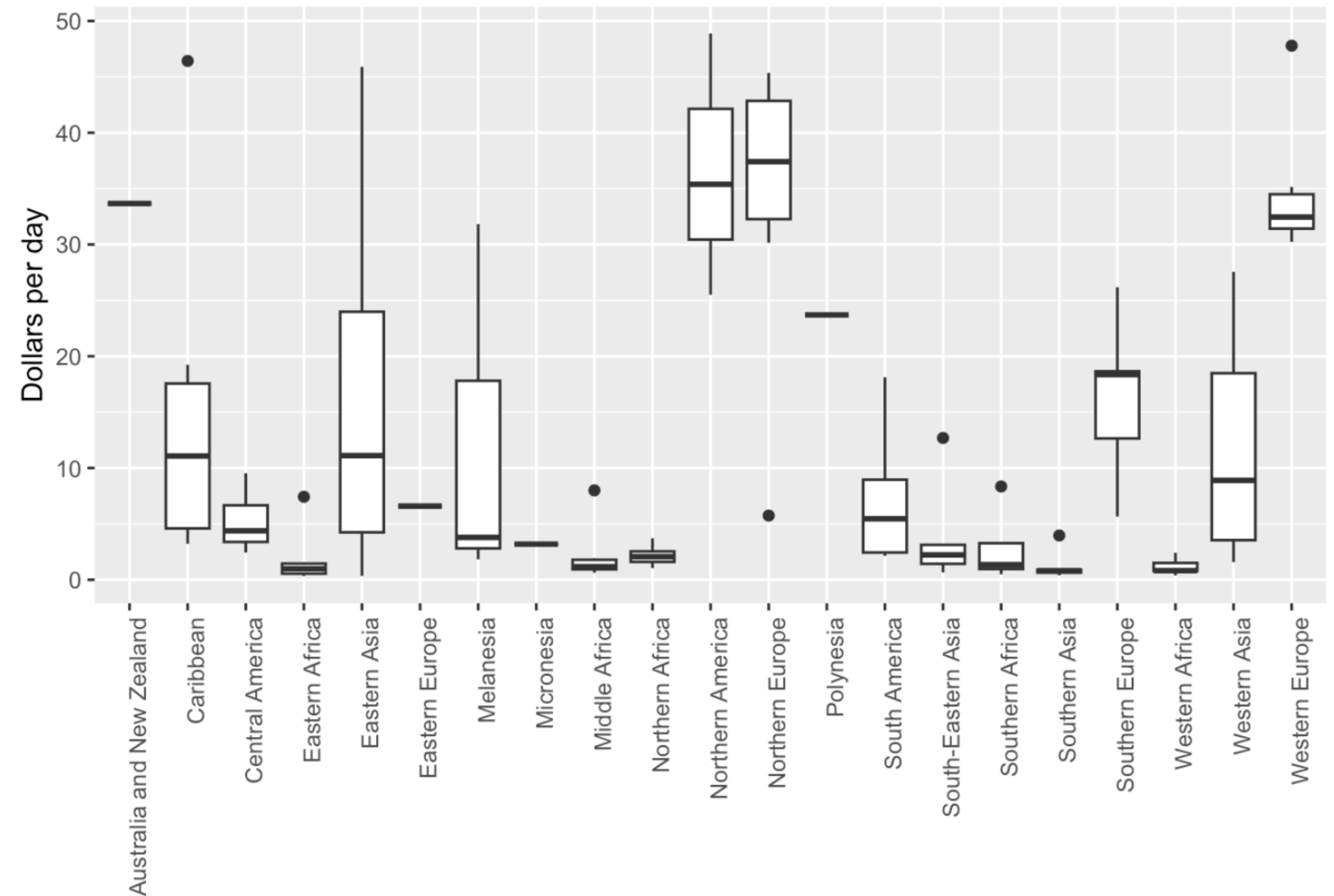
```
## [1] 22
```

```
gapminder %>%
  filter(year == past_year & !is.na(gdp)) %>%
  ggplot(aes(region, dollars_per_day)) +
  geom_boxplot() +
  xlab("Region") +
  ylab("Dollars per day")
```



Vertical axis labels

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  ggplot(aes(region, dollars_per_day)) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  xlab("") +  
  ylab("Dollars per day")
```



reorder

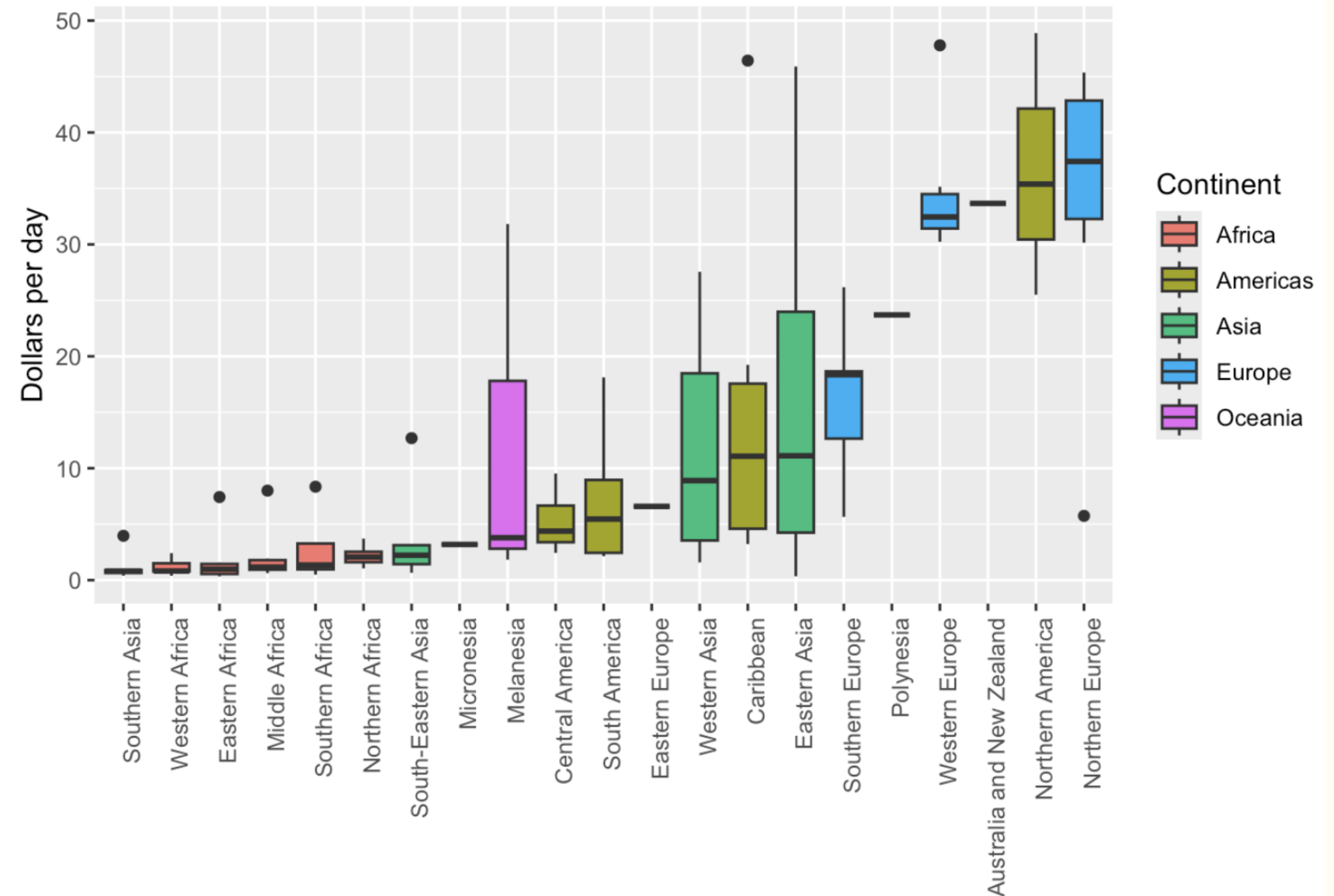
```
fac <- factor(c("Asia", "Asia", "West", "West", "West"))
levels(fac)
```

```
## [1] "Asia" "West"
```

```
value <- c(10, 11, 12, 6, 4)
fac <- reorder(fac, value, FUN = mean)
levels(fac)
```

```
## [1] "West" "Asia"
```

```
gapminder %>%
  filter(year == past_year & !is.na(gdp)) %>%
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%
  ggplot(aes(region, dollars_per_day, fill = continent)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  xlab("") +
  ylab("Dollars per day") +
  labs(fill = "Continent")
```



reorder

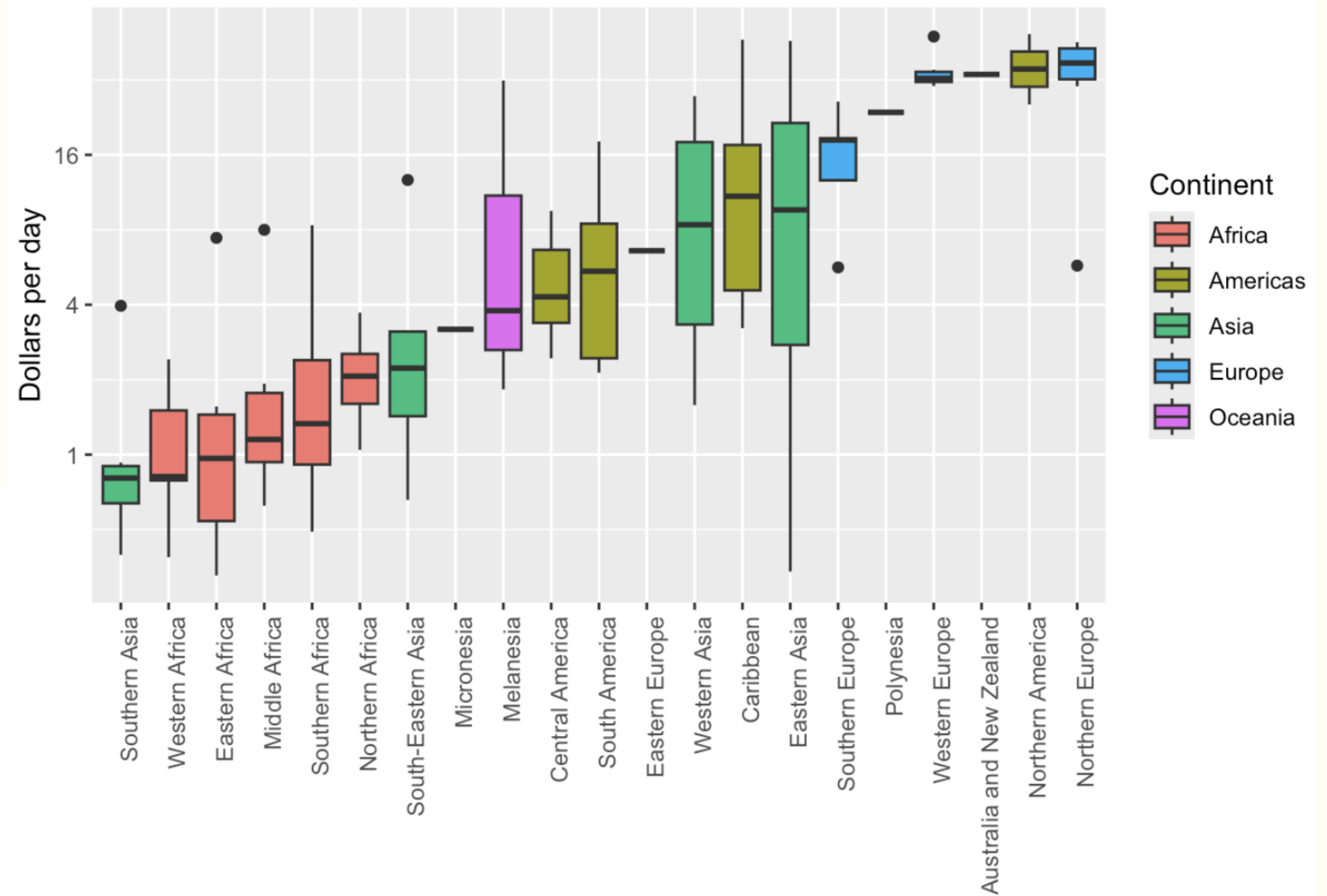
```
fac <- factor(c("Asia", "Asia", "West", "West", "West"))
levels(fac)
```

```
## [1] "Asia" "West"
```

```
value <- c(10, 11, 12, 6, 4)
fac <- reorder(fac, value, FUN = mean)
levels(fac)
```

```
## [1] "West" "Asia"
```

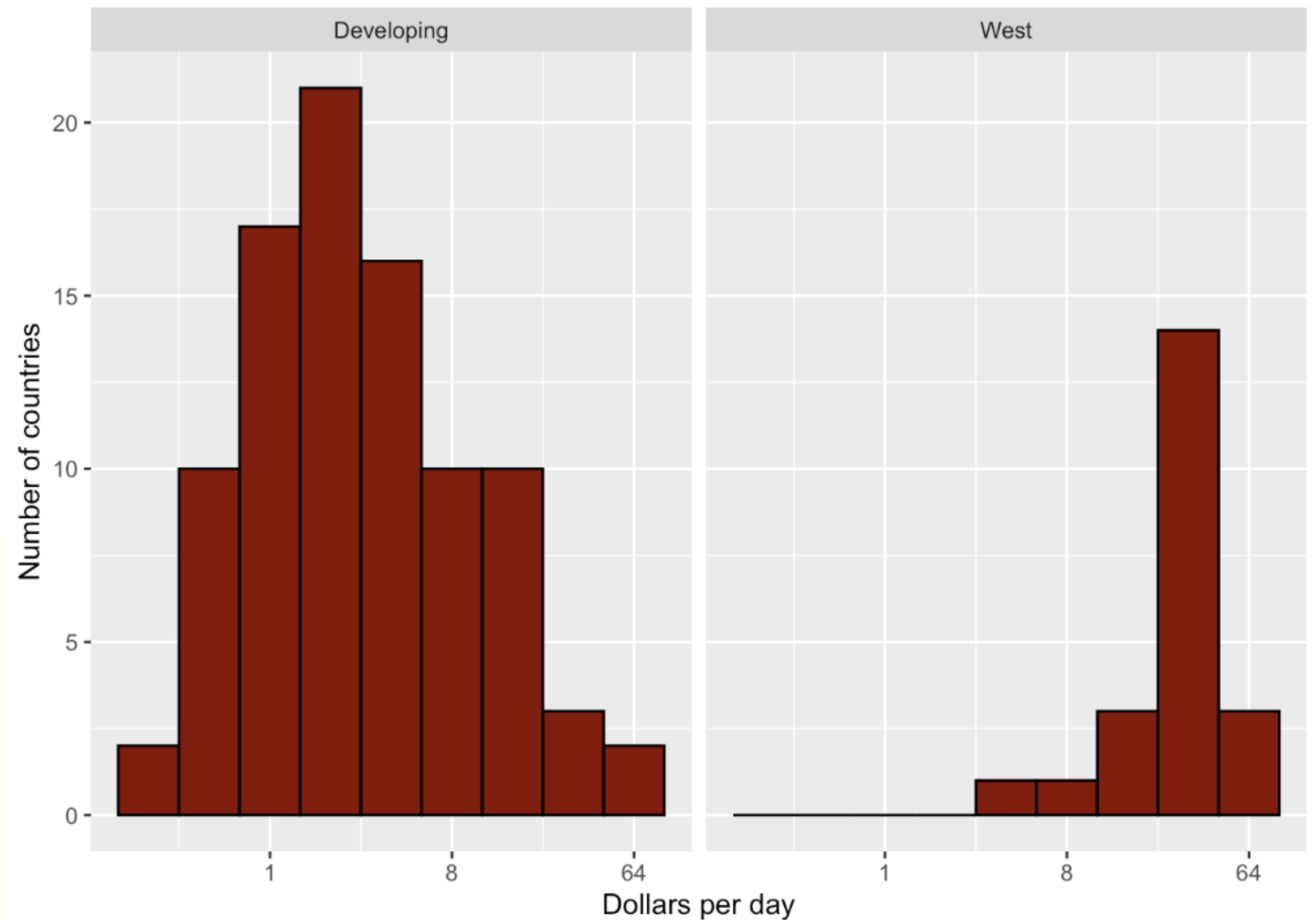
```
gapminder %>%
  filter(year == past_year & !is.na(gdp)) %>%
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%
  ggplot(aes(region, dollars_per_day, fill = continent)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  xlab("") +
  ylab("Dollars per day") +
  labs(fill = "Continent") +
  scale_y_continuous(trans = "log2")
```



Compare across years

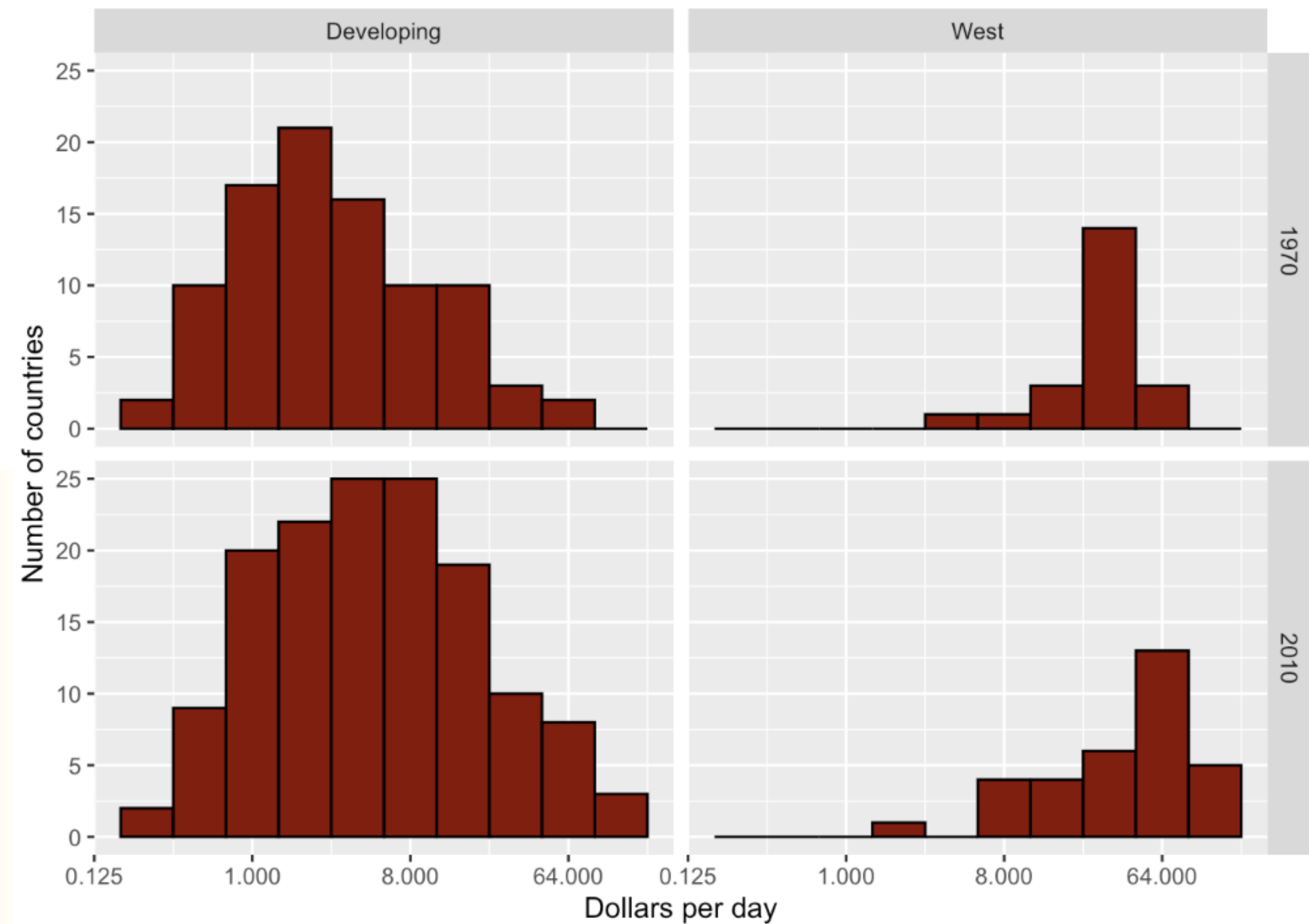
```
west <- c("Western Europe", "Northern Europe", "Southern Europe",  
         "Northern America", "Australia and New Zealand")
```

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_histogram(binwidth = 1, color = "black") +  
  scale_x_continuous(trans = "log2") +  
  facet_grid(. ~ group) +  
  xlab("Number of countries") +  
  ylab("Dollars per day")
```



Facet by two variables

```
past_year <- 1970
present_year <- 2010
gapminder %>%
  filter(year %in% c(past_year, present_year) & !is.na(gdp)) %>%
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%
  ggplot(aes(dollars_per_day)) +
  geom_histogram(binwidth = 1, color = "black") +
  scale_x_continuous(trans = "log2") +
  facet_grid(year ~ group) +
  xlab("Number of countries") +
  ylab("Dollars per day")
```



Make valid comparisons

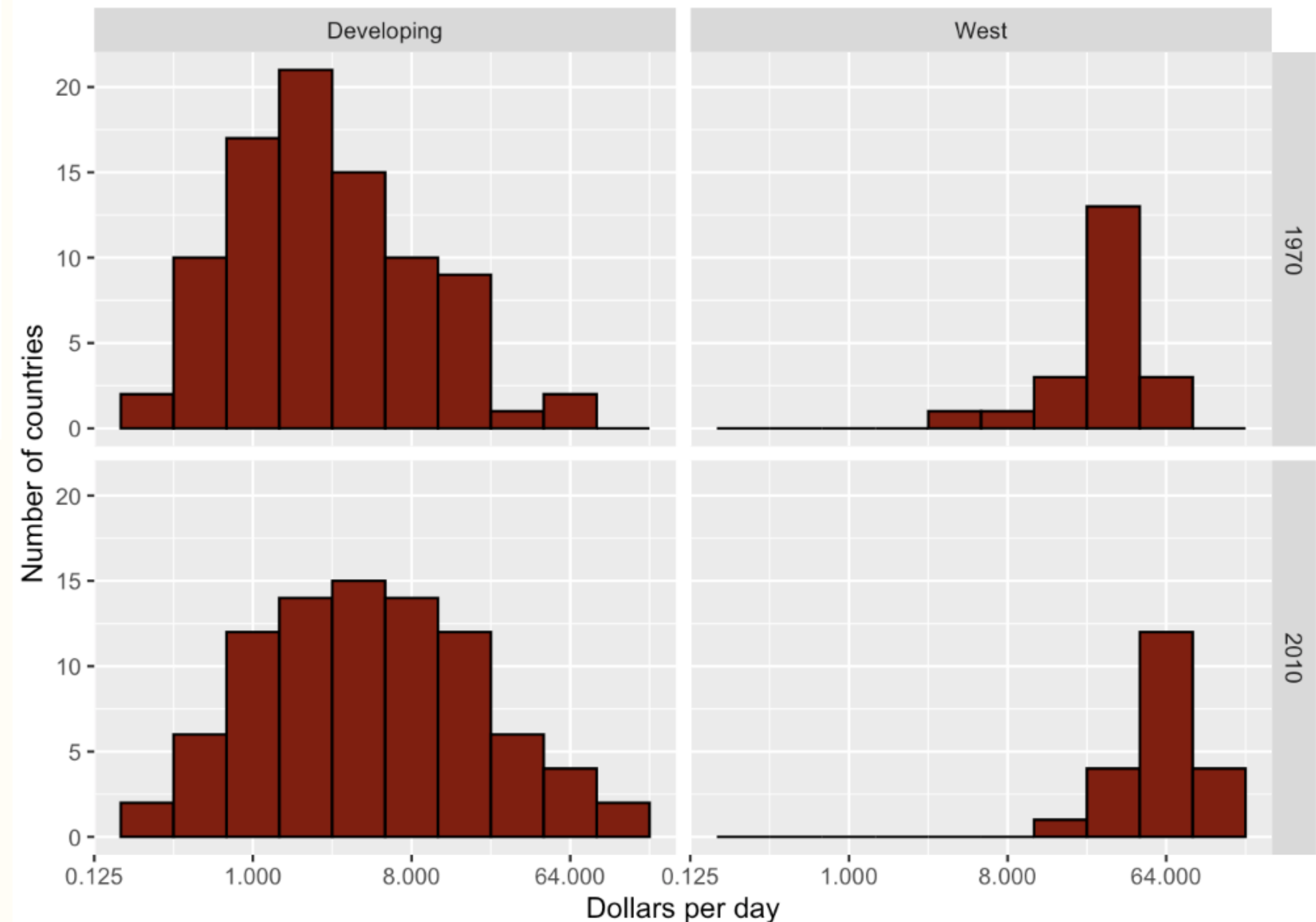
The number of countries represented in 1970 is less than the number in 2010

- More data for 2010
- Some countries split into multiple after 1970
- To make a **valid comparison** across years, we must have the **same countries for both years**

```
country_list_1 <- gapminder %>%  
  filter(year == past_year & !is.na(dollars_per_day)) %>%  
  .$country  
  
country_list_2 <- gapminder %>%  
  filter(year == present_year & !is.na(dollars_per_day)) %>%  
  .$country  
  
country_list <- intersect(country_list_1, country_list_2)
```

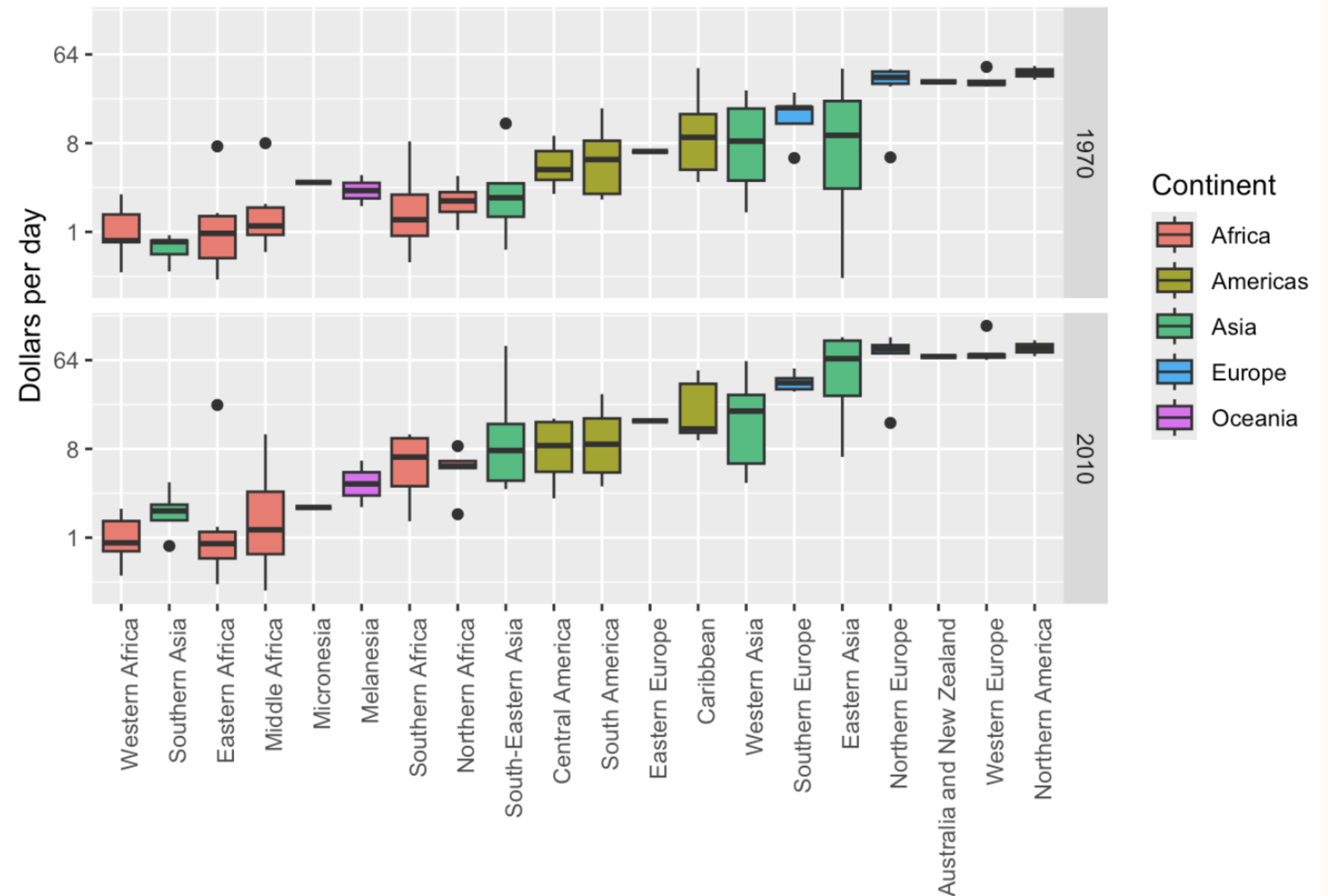
Make valid comparisons

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_histogram(binwidth = 1, color = "black") +  
  scale_x_continuous(trans = "log2") +  
  facet_grid(year ~ group) +  
  xlab("Number of countries") +  
  ylab("Dollars per day")
```



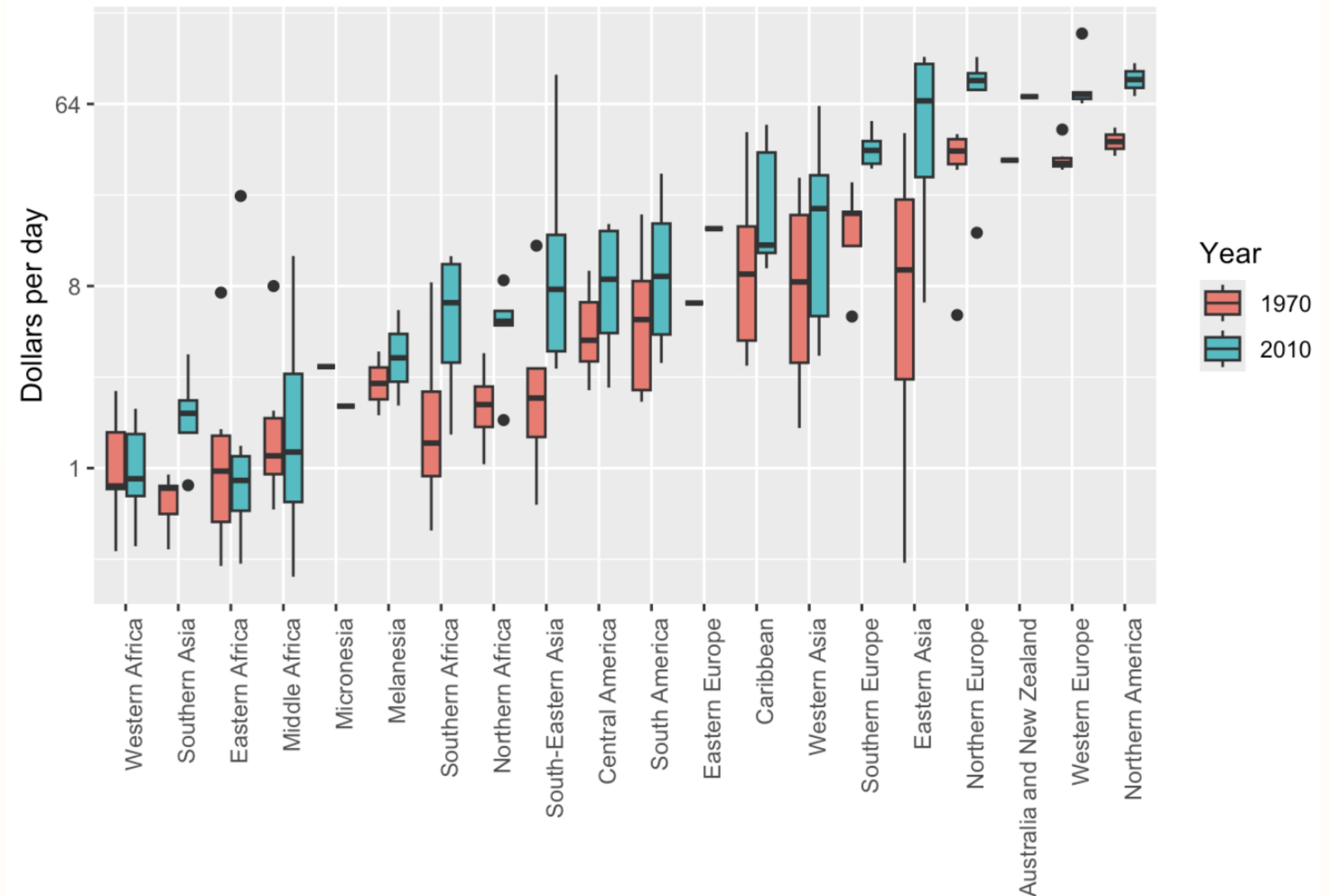
Boxplots across years

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%  
  ggplot(aes(region, dollars_per_day, fill = continent)) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  xlab("") +  
  ylab("Dollars per day") +  
  scale_y_continuous(trans = "log2") +  
  labs(fill = "Continent") +  
  facet_grid(year~.)
```



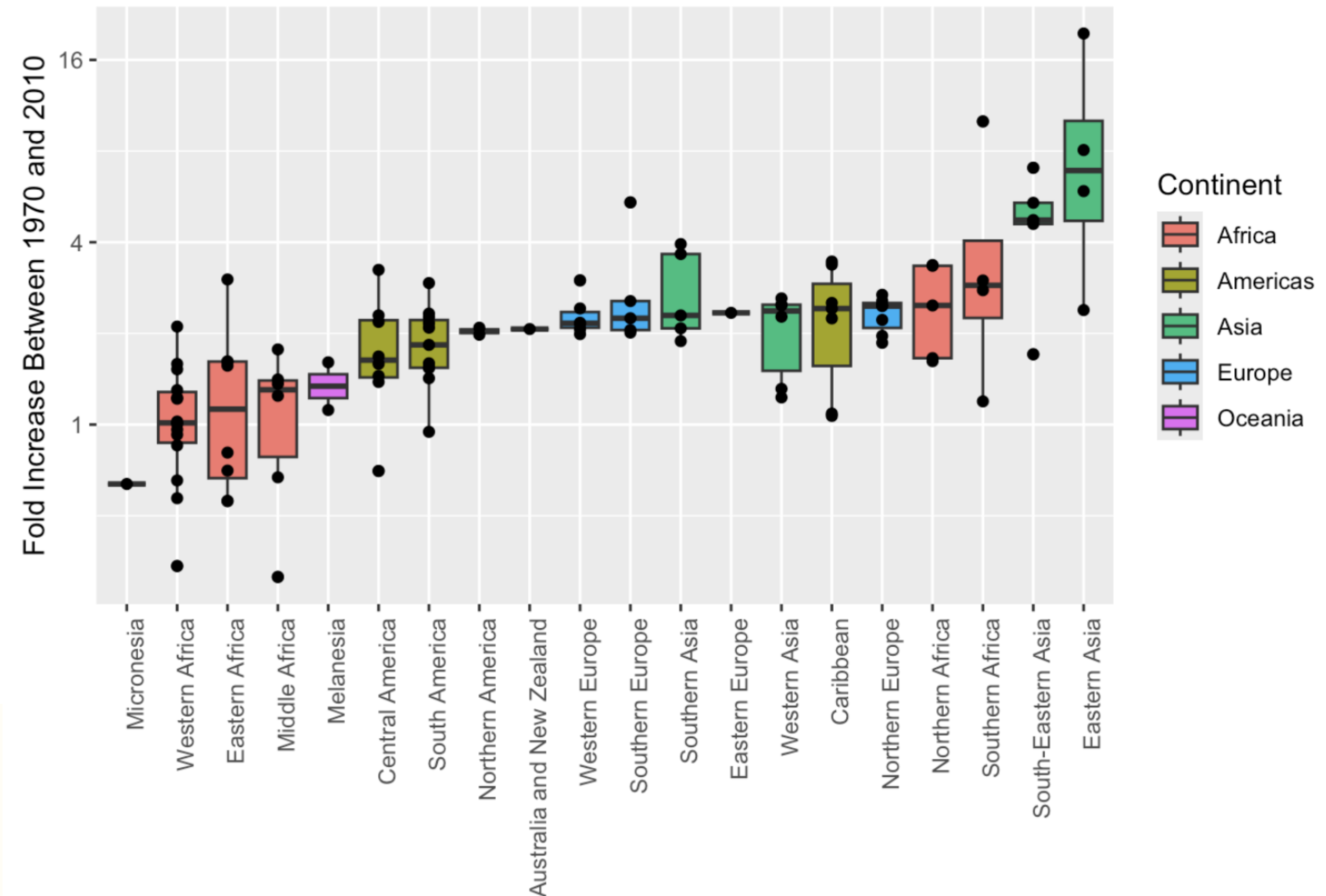
Plot side-by-side

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%  
  ggplot(aes(region, dollars_per_day, fill = factor(year))) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  xlab("") +  
  ylab("Dollars per day") +  
  scale_y_continuous(trans = "log2") +  
  labs(fill = "Year")
```



Fold change

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(year = ifelse(year == past_year, "past", "present")) %>%
  select(country, region, continent, year, dollars_per_day) %>%
  spread(year, dollars_per_day) %>%
  mutate(difference = present/past) %>%
  mutate(region = reorder(region, difference, FUN = median)) %>%
  ggplot(aes(region, difference, fill = factor(continent))) +
  geom_boxplot() +
  geom_point(show.legend = FALSE) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  xlab("") +
  ylab("Fold Increase Between 1970 and 2010") +
  labs(fill = "Continent") +
  scale_y_continuous(trans = "log2")
```



Summary

- Reordering by a meaningful value eases interpretation
- Comparisons of groups over time are only valid if the groups remain the same over time
- When comparing boxplots, plotting side-by-side rather than using faceting eases interpretation

Comparing distributions

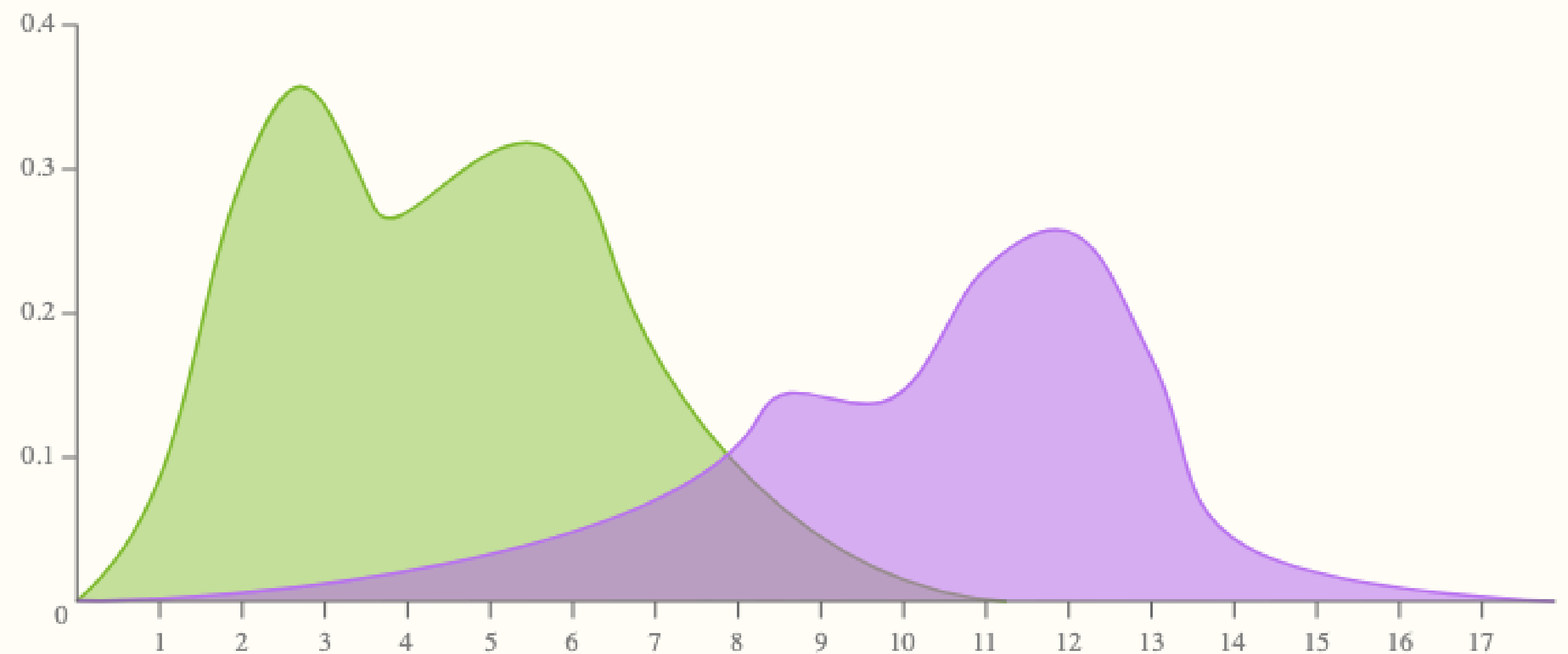
Smooth density plots

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Smooth density plots

Smooth density plots estimate the probability distribution of a continuous variable

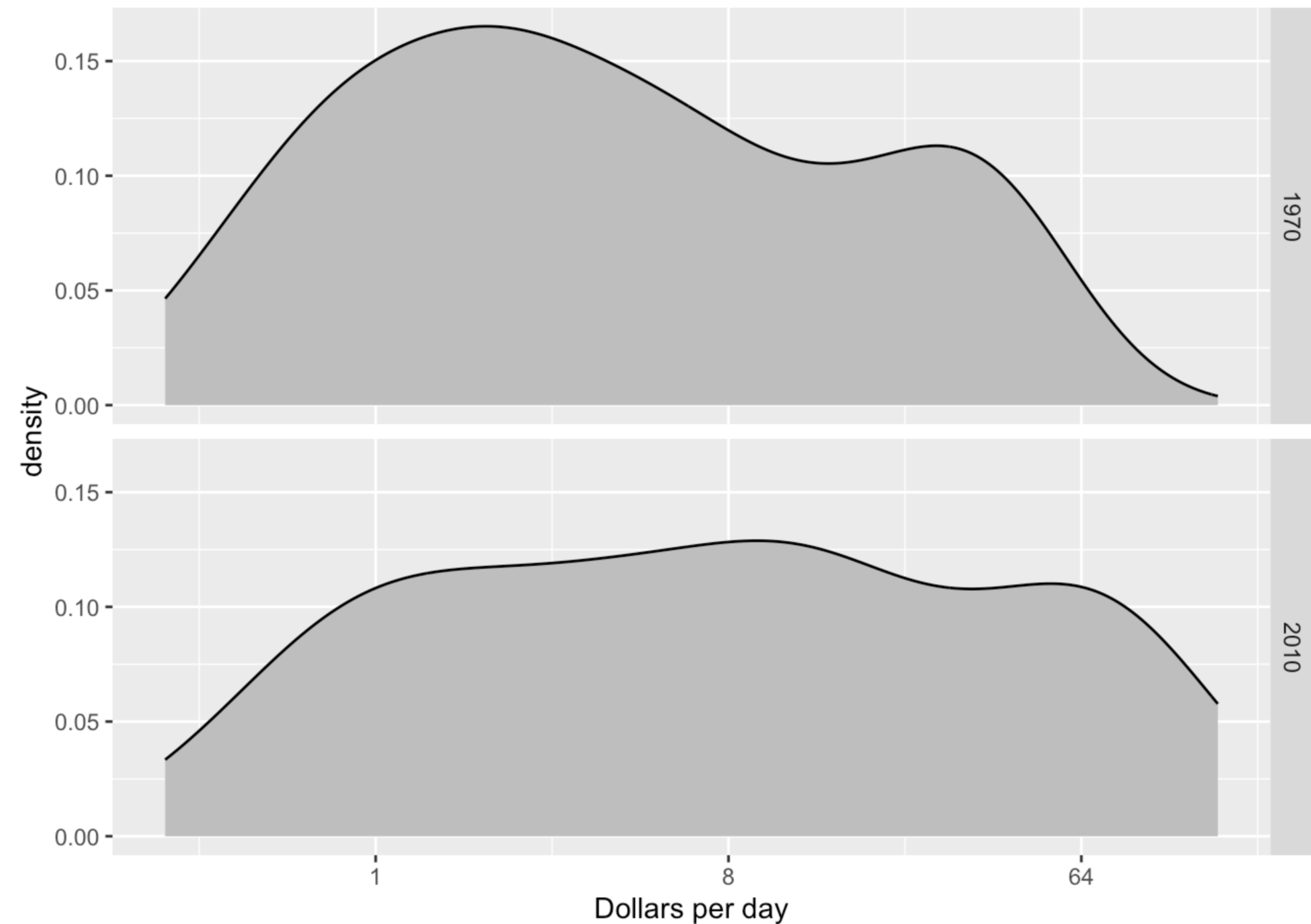
- A smoothed version of a histogram
- Useful for visualizing distribution shape
- Useful for comparing groups



Income distributions

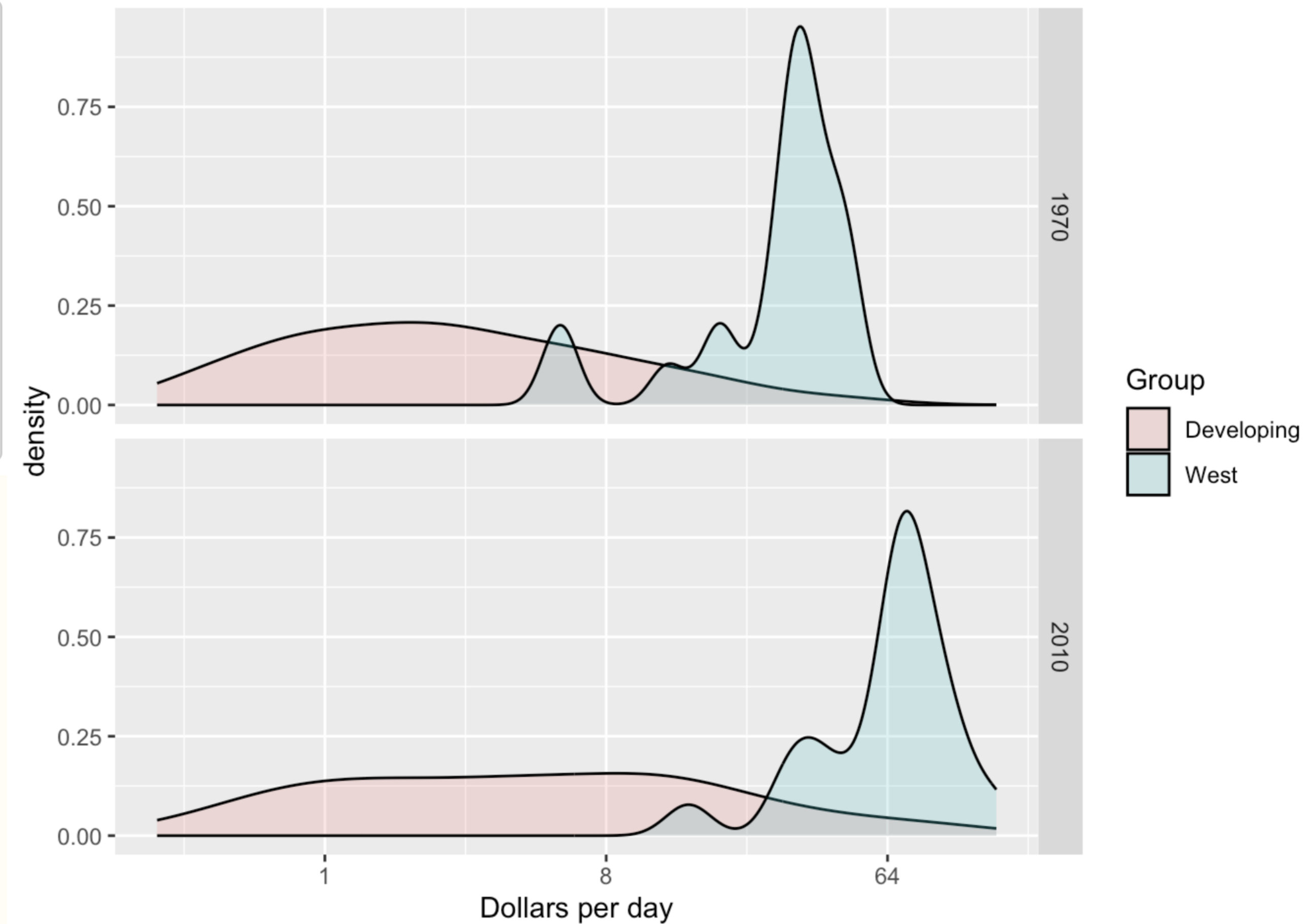
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
          country %in% country_list) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_density(fill = "grey") +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  facet_grid(year~.)
```

With this plot we can see that some countries have shifted to the right, closing the income gap – but we don't know **which countries** these are



Groups

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2) +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

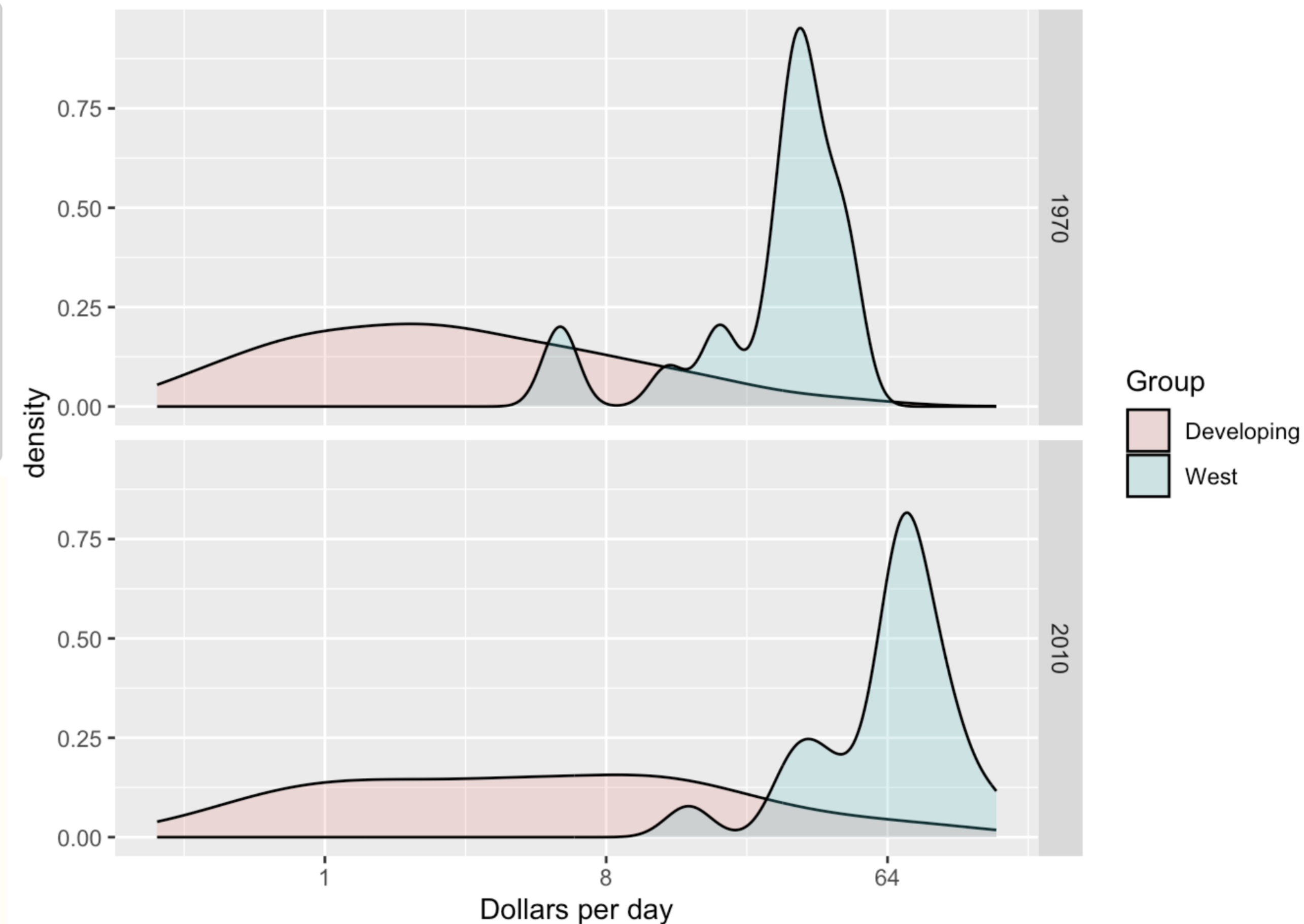


Groups

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%
  ggplot(aes(dollars_per_day, fill = group)) +
  scale_x_continuous(trans = "log2") +
  geom_density(alpha = 0.2) +
  facet_grid(year ~ .) +
  xlab("Dollars per day") +
  labs(fill = "Group")
```

The default is for the area under a smooth density plot to be **equal to 1**

This is not ideal in this context since there are 21 countries in our West group and 87 in our Developing group

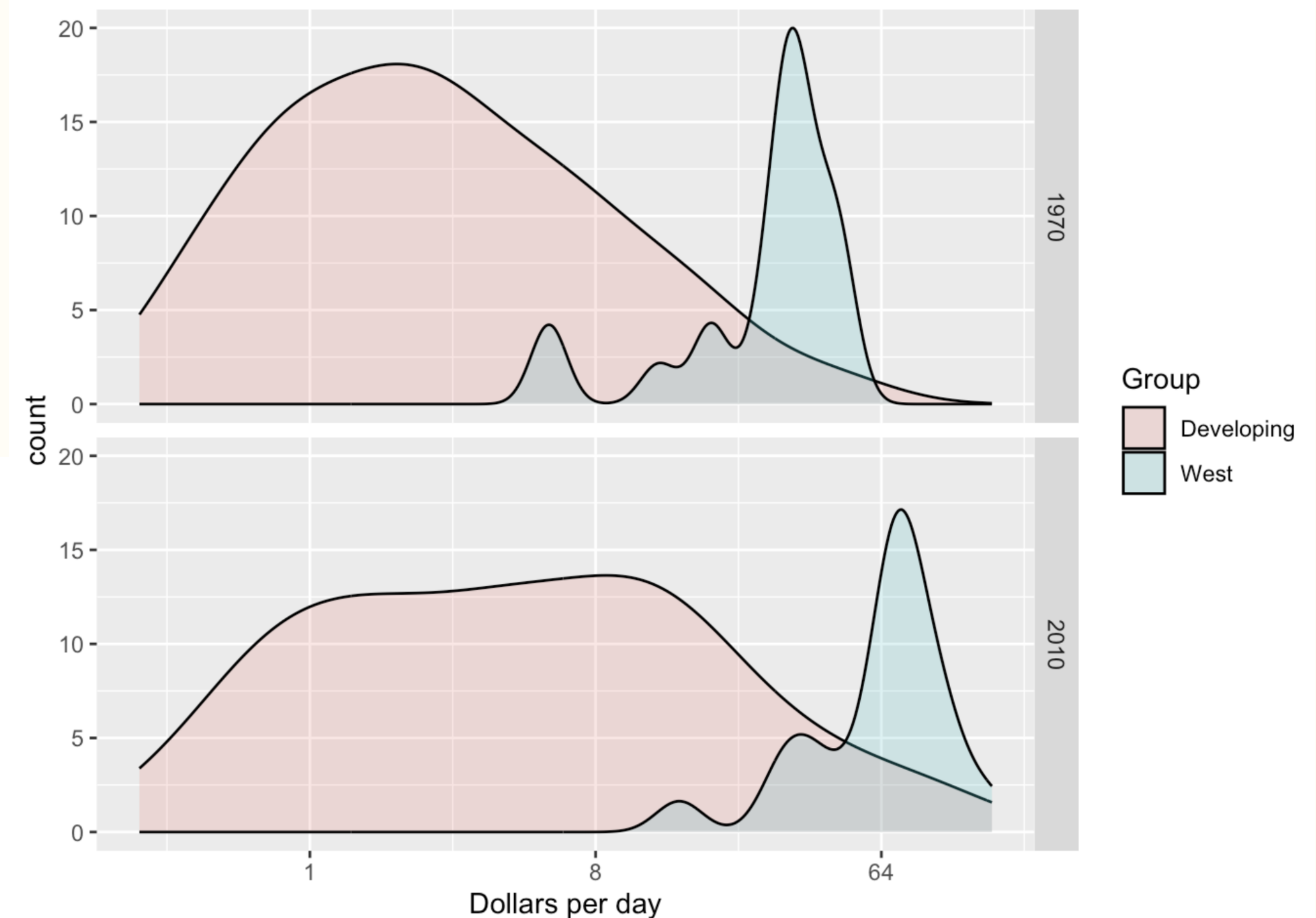


Accessing computed variables

Multiplying the y-axis values by the size of a group makes the area proportional to the number of countries in that group

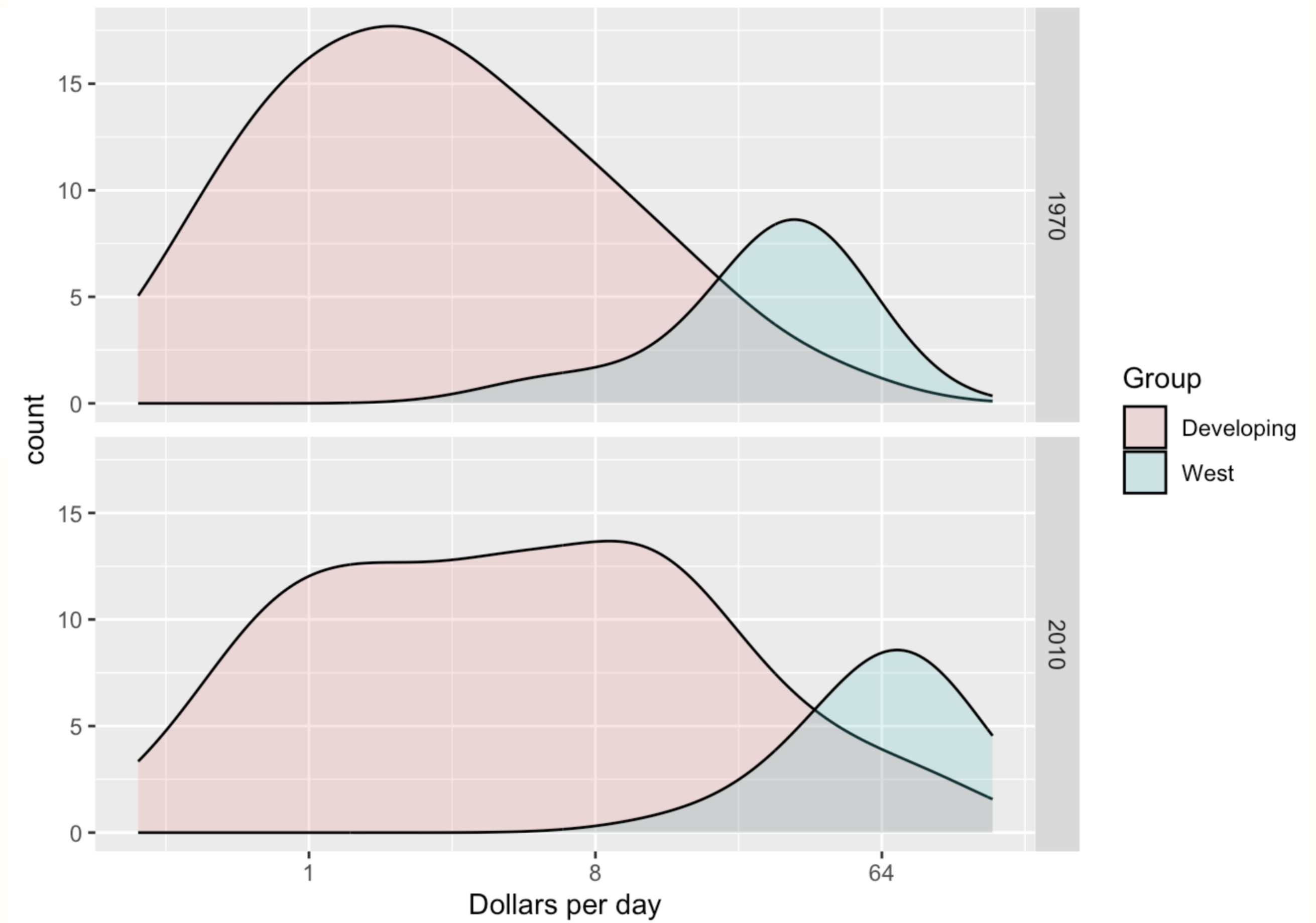
- **y = after_stat(count)**

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count), fill = group)) +  
  geom_density(alpha = 0.2) +  
  facet_grid(year ~ .) +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



Smoother densities (**bw**)

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%
  ggplot(aes(dollars_per_day, y = after_stat(count), fill = group)) +
  geom_density(alpha = 0.2, bw = 0.75) +
  facet_grid(year ~ .) +
  scale_x_continuous(trans = "log2") +
  xlab("Dollars per day") +
  labs(fill = "Group")
```



case_when

We can make this plot more informative by having more group categories

- The **case_when** function can create a new group variable efficiently

```
gapminder <- gapminder %>%  
  mutate(group = case_when(  
    region %in% west ~ "West",  
    region %in% c("Eastern Asia", "South-Eastern Asia") ~ "East Asia",  
    region %in% c("Caribbean", "Central America", "South America") ~ "Latin America",  
    continent == "Africa" & region != "Northern Africa" ~ "Sub-Saharan Africa",  
    TRUE ~ "Others"))
```

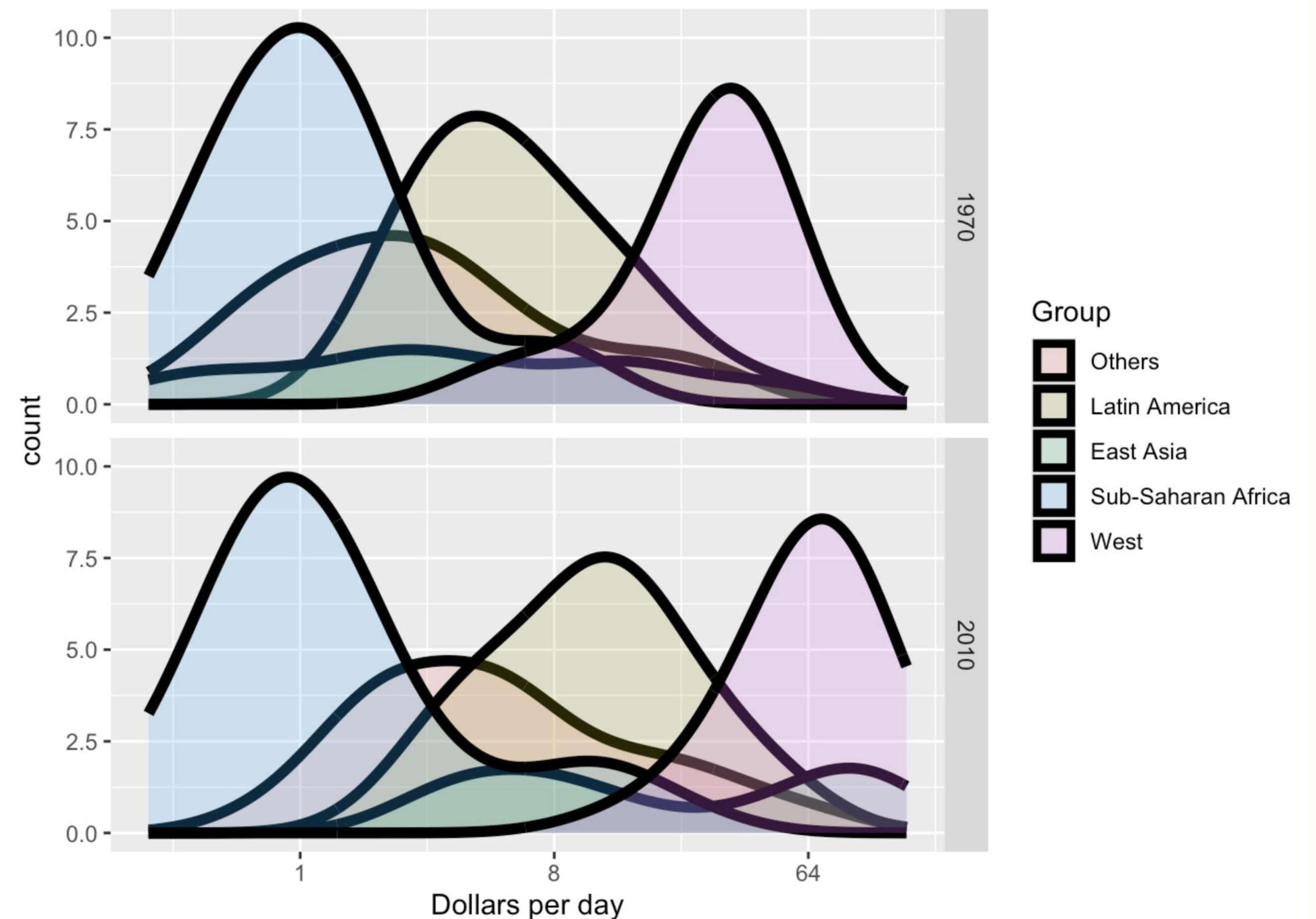
```
gapminder %>% select(region, group) %>% head()
```

##	region	group
## 1	Southern Europe	West
## 2	Northern Africa	Others
## 3	Middle Africa	Sub-Saharan Africa
## 4	Caribbean	Latin America
## 5	South America	Latin America
## 6	Western Asia	Others

linewidth

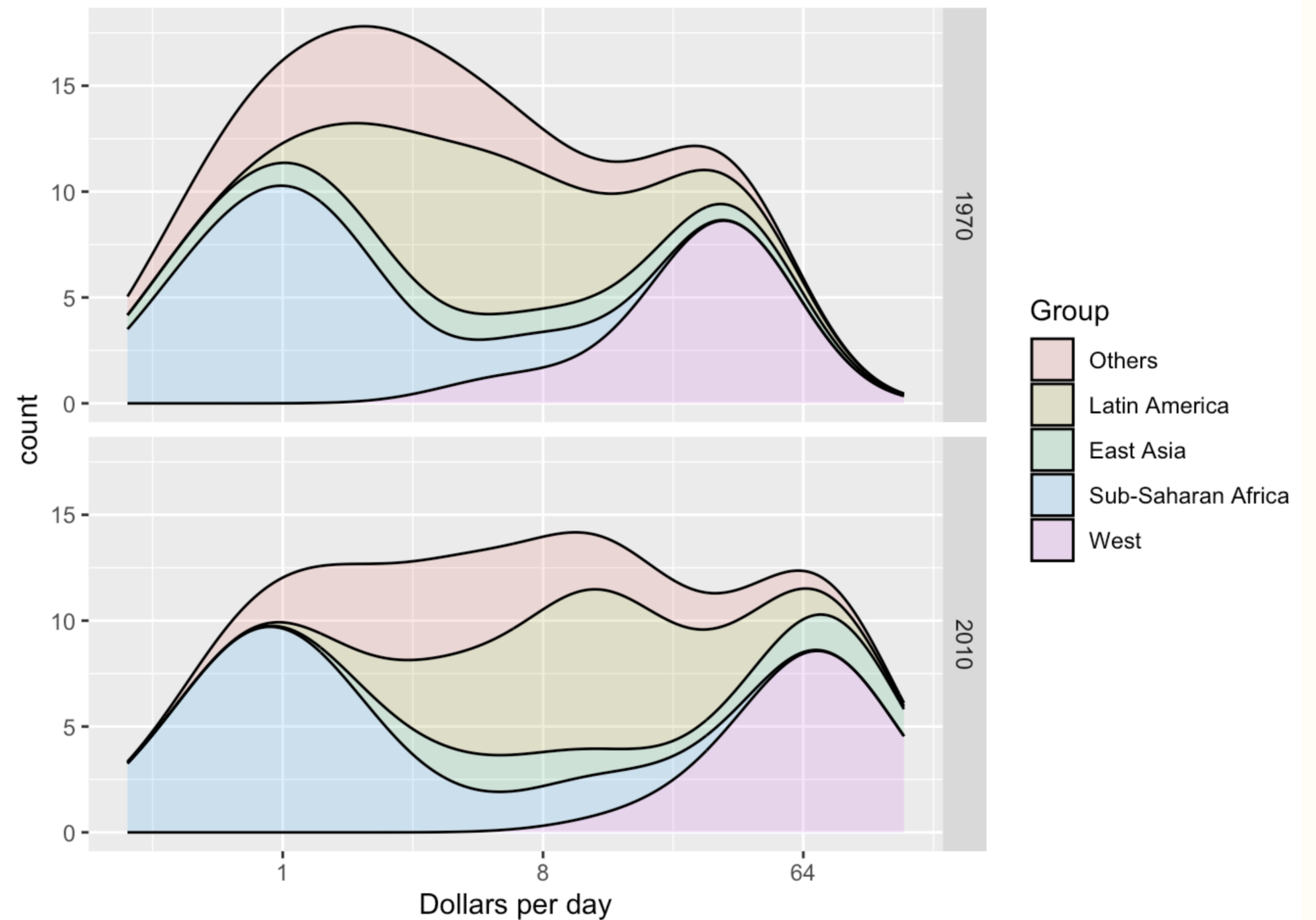
```
gapminder <- gapminder %>%  
  mutate(group = factor(group,  
    levels = c("Others", "Latin America",  
              "East Asia", "Sub-Saharan Africa",  
              "West")))
```

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
    country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
    fill = group)) +  
    scale_x_continuous(trans = "log2") +  
    geom_density(alpha = 0.2, bw = 0.75, linewidth = 2) +  
    facet_grid(year ~ .) +  
    xlab("Dollars per day") +  
    labs(fill = "Group")
```



position = "stack"

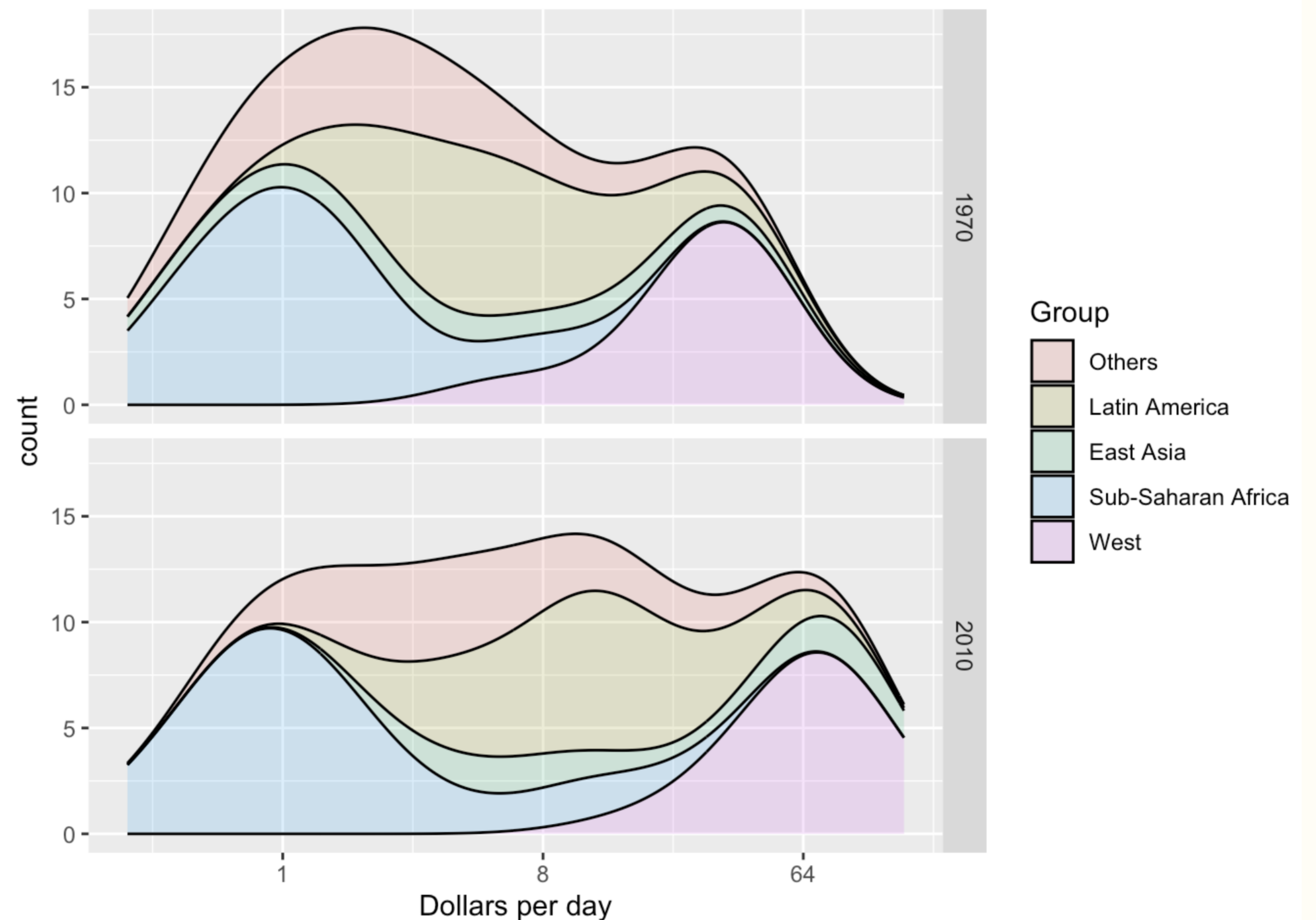
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



position = "stack"

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

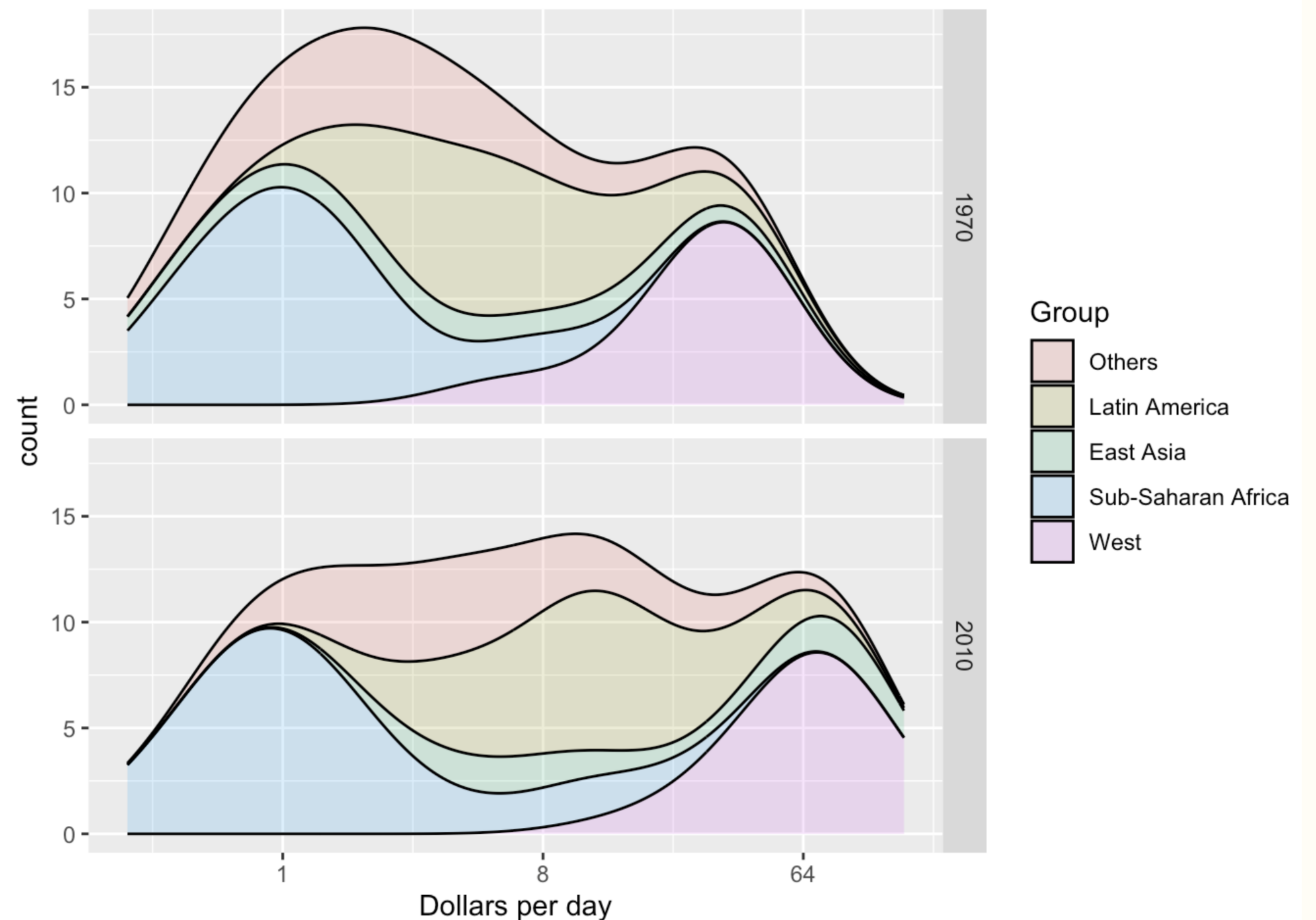
Plotting the two extremes (Sub-Saharan Africa and West) first shows the bimodality better



position = "stack"

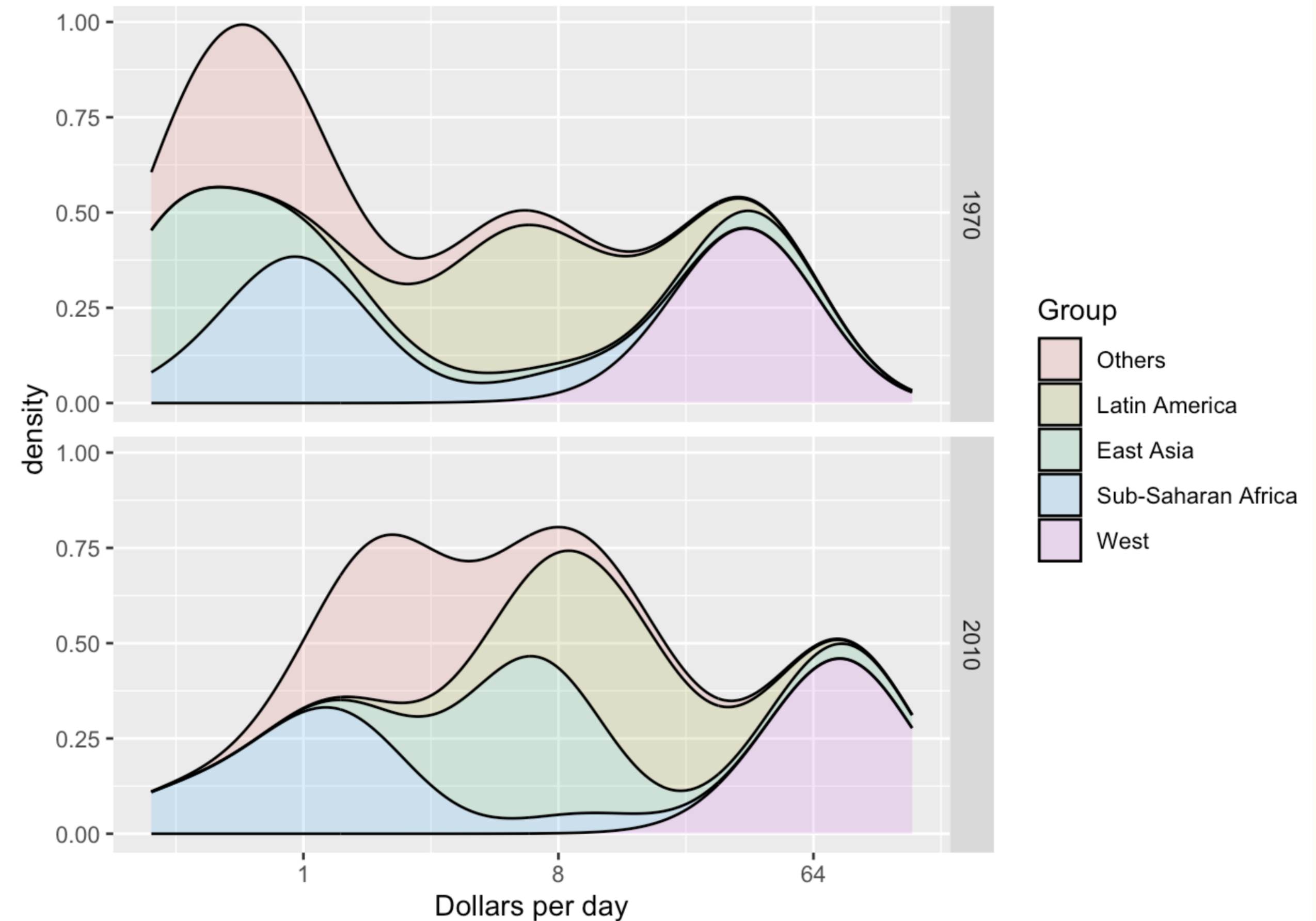
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

Here, each country has the same weight, which can skew comparisons – it's better to weight countries according to population



Weighted densities (**weight**)

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  group_by(year) %>%
  mutate(weight = population/sum(population)) %>%
  ungroup() %>%
  ggplot(aes(dollars_per_day, fill = group, weight = weight)) +
  scale_x_continuous(trans = "log2") +
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +
  facet_grid(year ~ .) +
  xlab("Dollars per day") +
  labs(fill = "Group")
```



Summary

- Distributions across groups can be compared using smooth density plots
- The **case_when** function is an efficient way to create a categorical (factor) variable
- Smooth density plots can be stacked using the **position = "stack"** argument
- Smooth density plots can be weighted using the **weight** argument